Internship Report

DEPARTMENT OF
COMPUTER ENGINEERING (REGIONAL LANGUAGE)

PCET’s Pimpri Chinchwad College of Engineering, Pune

Internship report of work in **Eduskills** done by
Piyush Hemant Hole

During the period of **April – June 2024** under the guidance of
Mrs. Rohini Sarode

AN INTERNSHIP REPORT ON

"Eduskills: Android Developer Virtual Internship"

SUBMITTED TO THE PIMPRI CHINCHWAD COLLEGE OF ENGINEERING, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE COURSE INTERNSHIP

OF

BACHELOR OF TECHNOLOGY
(COMPUTER ENGINEERING REGIONAL LANGUAGE)

SUBMITTED BY

PIYUSH HEMANT HOLE

DEPARTMENT OF COMPUTER ENGINEERING REGIONAL LANGUAGE

PCET'S PIMPRI CHINCHWAD COLLEGE OF ENGINEERING,

NIGDI, PUNE 411044

(Academic Year 2024-25)

CERTIFICATE

This is to certify that "**Piyush Hole"** has successfully completed his internship work on **"Android Developer Virtual Internship"** at "**Eduskills"** during the "**Internship period"** in partial fulfillment for the award of the **Engineering Degree in Computer Engineering (Regional Language)**.

Mrs. Rohini Sarode
Name of guide (Internal)

Dr. Rachana Y. Patil
Head, Department of Computer Engineering (Regional Language)

Place: Pune

Date: 09/4/2025

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Chapter 1: Introduction

The **Eduskills Android Developer Virtual Internship**, in collaboration with **Google for Developers** and supported by the **All-India Council for Technical Education (AICTE)**, was conducted from **April to June 2024**. This initiative aimed to bridge the gap between academia and industry by equipping students with practical skills in **Android app development using Kotlin**

During the internship, participants were introduced to **the fundamentals of Android development**, including setting up the **Android Studio development environment**, understanding **Kotlin programming basics**, and implementing essential **Android app components**. The program followed a structured learning approach, where interns had to complete and submit various tasks at each stage of learning.

As a final project, interns were required to **develop a fully functional Android application**. Since this internship was completed alongside the **Open Elective course on Android Development using Kotlin**, the knowledge gained was reinforced through both practical implementation and academic learning. The project developed during the internship was an **Android Expense Tracker App**, integrating key Android development concepts.

Internship Discussion

The **Eduskills Android Developer Virtual Internship** followed a structured approach, starting from the **basics of Android development using Kotlin** and progressively moving towards building a full-fledged Android application.

We began by **understanding the fundamentals of Android development**, including the **Android architecture, key components, and UI design principles**. The next step involved **setting up the development environment**, where we learned how to install and configure **Android Studio, SDKs, and necessary dependencies**.

As the internship progressed, we were assigned **hands-on tasks** to practice Kotlin programming, build simple UI layouts, and implement core Android components like **Activities, Fragments, RecyclerView, ViewModel, and LiveData**. These tasks helped reinforce our learning and provided a solid foundation for advanced concepts.

To apply all the concepts we had learned, we worked on a **final project – an Android Expense Tracker App**. This project integrated key features like **adding and managing transactions, categorizing expenses, displaying financial summaries, and using a Room Database for data storage**. The app also implemented **real-time UI updates using LiveData and ViewModel**, ensuring a smooth user experience.

By the end of the internship, we had gained **practical experience in mobile app development**, strengthened our **problem-solving skills**, and understood **best practices in Android development**. This hands-on learning approach provided a comprehensive understanding of **Android app development using Kotlin**, bridging the gap between theoretical knowledge and real-world implementation.

Problem Statement:

Managing personal finances effectively is a common struggle faced by individuals. Many users find it difficult to keep track of their income and expenses, leading to poor financial management and oversight of their spending habits. Existing solutions, while helpful, are often too complex or lack the user-centered design needed for everyday financial tracking.

The Expense Tracker App addresses this issue by providing a simple yet powerful tool that helps users record and categorize their financial transactions with ease. The app aims to present a clear overview of their financial health by displaying total balance, income, and expenses on the home screen, while also offering personalized insights into their spending patterns through dynamic statistics.

The problem it solves is the lack of an intuitive, easy-to-use, and real-time financial management tool that fits seamlessly into users' daily routines, enabling better control over their financial well-being.

Project Objectives:

- Track Income and Expenses: Enable users to add and categorize their income and expenses with details like source, amount, and date for effective financial tracking.

- Display Overall Balance: Provide a clear, up-to-date overview of the total balance, showing both income and expense details on the home screen.

- Dynamic Transaction List: Show a list of recent and past transactions with sorting and filtering options for easy navigation through the transaction history.

- Visualize Expense Trends: Offer graphical representation through a dynamic statistics screen to help users understand their spending habits and trends over time.

- Localized Currency: Automatically detect and use the appropriate currency format based on the user's location.

- User-Friendly Interface: Maintain an intuitive, simple, and visually appealing user interface to ensure that financial management is easy and accessible

Motivation:

- **Efficient Financial Management:** The need for an intuitive platform that allows users to manage their finances without the complexities found in existing apps like Mint or YNAB.

- **User-Friendly Interface:** Current financial apps often overwhelm users with too many features. The motivation was to build a simple yet effective app that provides financial clarity with minimal effort.

- **Real-Time Insights:** Users require immediate updates on their income, expenses, and overall financial balance, helping them make informed financial decisions.

- **Enhanced Personalization:** By incorporating dynamic greetings and personalized statistics, the app aims to create a more engaging and user-centric experience.

- **Local Data Management:** Many users prefer local data storage for privacy and security reasons, prompting the decision to use the **Room Database** for storing transaction data locally

Scope:

Core Functionality: The app allows users to manage their personal finances by adding, categorizing, and viewing income and expense transactions.

Transaction History: Users can view, filter, and sort transactions by date and type, providing an organized transaction history for financial review.

Real-Time Balance Update: Displays the user’s current financial status, including total balance, income, and expenses on the home screen.

Local Data Storage: All transaction data is stored locally using the **Room Database**, ensuring quick access and privacy without relying on external servers or cloud storage.

Statistics Visualization: Users can view dynamic graphs representing their spending trends over time, helping them better understand their financial habits.

Modular Design: The app is structured into separate sections (Home, Add Transaction, Transaction List, and Statistics), making navigation seamless and intuitive.

Limitations: The current scope does not include user authentication, transaction editing, or cloud backup, though these features are planned for future updates

Chapter 2: Requirement Analysis

Functional Requirements:

a) **Transaction Management:**
Users must be able to add transactions by specifying type (income or expense), category, amount, and date. The app will allow users to view, sort, and filter transactions based on type or date range. Future functionality will include editing and deleting transactions.

b) **Home Screen Display:**
The home screen will display real-time updates of total balance, income, and expenses. It should also feature quick-access buttons for adding new transactions efficiently.

c) **Statistics Screen:**
A dynamic statistics screen will display a graphical representation of user spending trends over time. The graph must update instantly when new transactions are added.

d) **Localization:**
The app will detect the user’s location and automatically display the appropriate currency symbol (e.g., ₹ for India).

e) **Data Storage:**
The app will use the Room Database for local data storage, enabling efficient CRUD (Create, Read, Update, Delete) operations for transaction records.

f) **Filters and Sorting:**
Users will be able to filter transactions by type (income/expense) and date range, providing an easy way to navigate their transaction history.

g) **Dynamic Greetings:**
The home screen will display a greeting message that dynamically changes based on the time of day (e.g., "Good Morning," "Good Afternoon").

Non-Functional Requirements:

a) **Performance:**
The app must load transaction history and financial summaries quickly (within 2 seconds). Dynamic graphs should update in real-time with minimal delay after transactions are added.

b) **Usability:**
The app must offer an intuitive, user-friendly interface with seamless navigation between screens. Error messages should clearly guide users in case of incorrect inputs or system errors.

c) **Reliability:**
The app must ensure the local storage of transactions without data loss, even if the app crashes or the device restarts. A future update will include cloud storage for secure backups.

d) **Security:**
While the current version does not include user authentication, future updates will incorporate login security and encryption for storing sensitive financial data.

e) **Scalability:**
The app must handle large volumes of transaction data efficiently without compromising performance.

f) **Maintainability:**
The app follows the MVVM (Model-View-View Model) architecture, allowing easy updates and the integration of future features or fixes.

g) **Testability:**
Functional requirements will be tested through unit tests, particularly for adding and filtering transactions. UI testing will ensure responsiveness and compatibility across various Android device sizes.

Chapter 3: Project Design

System Design:

The design phase of the Expense Tracker app was centered around creating a simple, intuitive, and user-friendly experience. The app's interface was structured to provide essential features without overwhelming the user. The key aspects of the design methodology include:

- **User-Centered Interface:** Prioritizes ease of navigation with a clear layout that highlights relevant information.
- **Modular Structure:** Divided into four main screens (Home, Add Transaction, Transaction List, Statistics), each serving a distinct purpose for easy navigation**.**
- **Dynamic Greetings:** Personalized greetings on the Home Screen change based on the time of day, enhancing user experience**.**
- **Clean UI with Focused Features:** Simplified design avoids unnecessary complexity, focusing on tracking income and expenses with intuitive buttons.
- **Seamless Navigation:** Easy access to add transactions, view history, and analyze spending with straightforward filtering options.
- **Visual Data Representation:** The Statistics screen features dynamic graphs that update with new transactions, providing clear insights into spending habits.

Database Design:

- Expense Table:
  - Table Name: expense_table
  - Attributes:
    - id (INTEGER, Primary Key, Auto Increment)
    - title (TEXT, Not Null)
    - amount (REAL, Not Null)
    - date (TEXT, Not Null)
    - type (TEXT, Not Null)
  - ExpenseSummary consisting Type, Date, Total_Amount

Backend Development:

- Implement the core logic to Manage expenses and income tracking functionality.
- Implement CRUD operations using Room database for ExpenseEntity.
- Implement data validation and exception handling.

Frontend Development:

- **User Interface Design**: Create intuitive layouts for Home, Add Transaction, Transaction List, and Statistics screens.
- **Compose UI Components**: Use Jetpack Compose to build responsive UI elements

like TextViews, Buttons, and LazyColumn.
  - **User Interaction Handling**: Implement click listeners and input validations for user actions, ensuring a smooth user experience

Development:

- **Technology**: Kotlin
- **Architecture**: MVVM pattern with Room Database for data storage
- **Key Android Components**:
  - **EditText**: For user input, such as entering transaction amounts.
  - **DropdownMenu (Spinner)**: For selecting income/expense categories and date ranges.
  - **TextView**: For displaying balances, transaction summaries, and transaction details.
  - **RecyclerView (LazyColumn in Jetpack Compose)**: For listing transactions with filters.
  - **GraphView (Chart Library)**: For visualizing expense trends on the statistics screen.
  - **ViewModel and LiveData**: For managing and observing UI data.
  - **Room**: For local database storage and CRUD operations on transactions.

App Features:

1. **Home Screen:**
   - Displays an overview of **Total Balance, Income, and Expenses**.
   - Users can view a list of recent transactions with their amounts and dates.
   - Buttons to add **New Income** or **New Expense**.

2. **Add Transaction Screen:**
   - Users can add transactions with the following details:
     - **Type:** Income or Expense.
     - **Amount:** The transaction amount.
     - **Date:** Transaction date.
     - **Category:** Select the source or category (e.g., Salary, Food, Entertainment).

3. **Transaction List Screen:**
   - Displays a full list of all transactions, sorted by date.
   - Users can filter transactions by:
     - **Type**: Income or Expense.
     - **Date Range**: Today, Yesterday, Last 30 days, Last Year, etc.
   - Swipe left on a transaction to reveal a delete option, allowing users to remove an entry.

4. **Statistics Screen:**
   - Dynamic graphs update in real-time based on the transaction data.
   - Visualize spending trends, categorized breakdowns, and overall financial health.

Chapter 4: Learning Experience

a) Knowledge Acquired

Throughout the **Eduskills Android Developer Virtual Internship**, I gained extensive knowledge in **Android app development using Kotlin**. The training covered fundamental concepts such as **Android architecture, UI design, activity lifecycle, fragments, ViewModel, LiveData, and Room Database**. These topics were closely aligned with the concepts covered in the **Open Elective course on Android Development using Kotlin**, reinforcing my classroom learning with hands-on experience.

Additionally, I learned about **dependency injection, state management techniques, debugging tools in Android Studio, and best practices for app performance optimization**. This knowledge not only enhanced my technical expertise but also helped me understand how professional Android applications are structured and built.

b) Skills Learned

The internship provided an opportunity to develop both **technical and professional skills**, including:

- **Mobile App Development** – Building a fully functional Android app from scratch.
- **Kotlin Programming** – Writing clean, efficient, and object-oriented code.
- **UI/UX Design Principles** – Creating user-friendly interfaces using **Jetpack Compose and XML layouts**.
- **State Management** – Implementing **ViewModel and LiveData** to manage UI updates efficiently.
- **Local Data Storage** – Using **Room Database (SQLite ORM)** to store and retrieve transaction data.
- **Debugging & Testing** – Using **Android Studio tools, Logcat, and breakpoints** to fix bugs and optimize performance.
- **Version Control (Git & GitHub)** – Managing project versions and collaborating effectively.

Beyond technical skills, I also improved my **problem-solving abilities, time management, and adaptability**. Completing tasks within deadlines while learning new concepts on the go was a valuable experience.

c) Observed Attitudes and Gained Values

The internship emphasized several key professional values essential for career success, including:

- **Consistency and Commitment** – Completing tasks systematically and meeting deadlines.
- **Attention to Detail** – Writing clean, maintainable code and ensuring UI consistency.
- **Adaptability and Continuous Learning** – Quickly learning new technologies and best practices.
- **Collaboration and Communication** – Engaging with mentors and peers to seek feedback and share ideas.

- **Problem-Solving Mindset** – Identifying issues, analyzing them systematically, and implementing effective solutions.

I also observed the importance of **professional ethics, patience, and perseverance**, which are crucial for success in the software industry.

d) The Most Challenging Task Performed

The most challenging task I encountered was **implementing real-time UI updates in the Expense Tracker App**. Initially, when transactions were added or deleted, the total balance and transaction list did not update automatically.

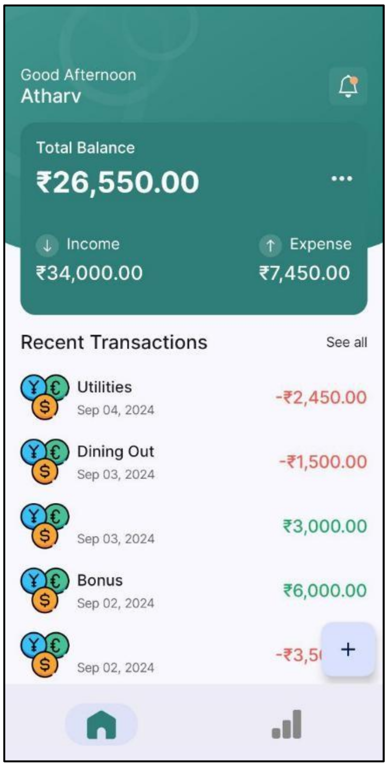
To resolve this issue, I researched and implemented **LiveData with ViewModel**, ensuring that UI components were automatically updated whenever data changed. This required a deep understanding of **Android's lifecycle-aware components**, data binding, and the observer pattern.

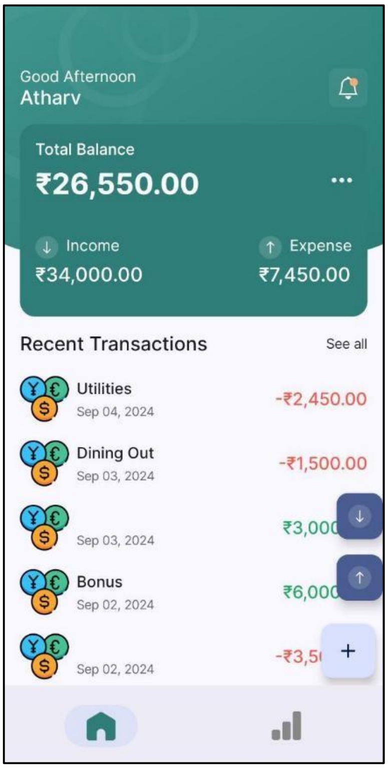
Despite the challenge, overcoming this issue significantly improved my problem-solving skills and reinforced my understanding of **MVVM architecture in Android development**. The experience taught me the importance of **debugging systematically, seeking solutions from documentation, and continuously improving code efficiency**.

Overall, this internship provided a **well-rounded learning experience**, equipping me with **practical Android development skills, professional work ethics, and the confidence to tackle real-world software development challenges**.

Chapter 5: Results (Screenshots)

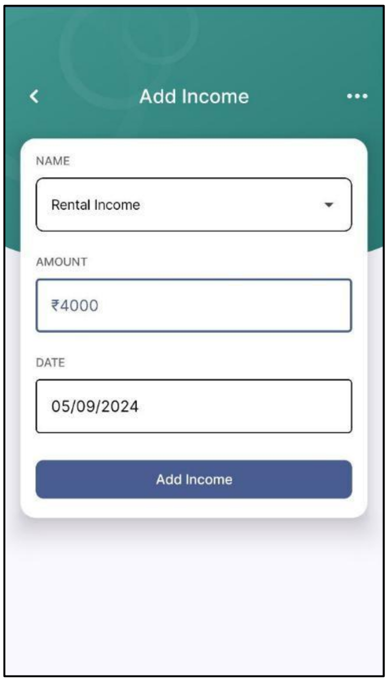
1. Home Screen:

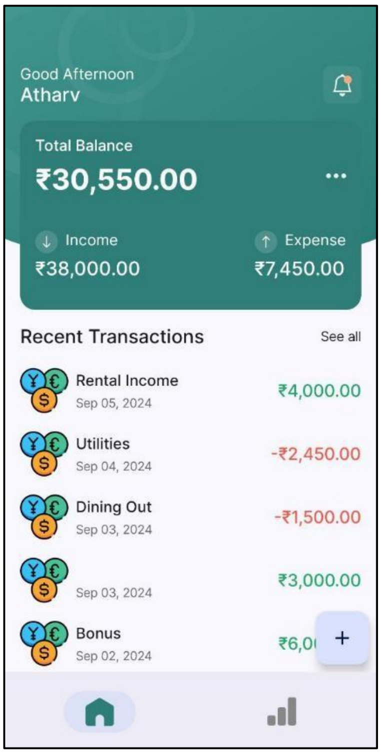




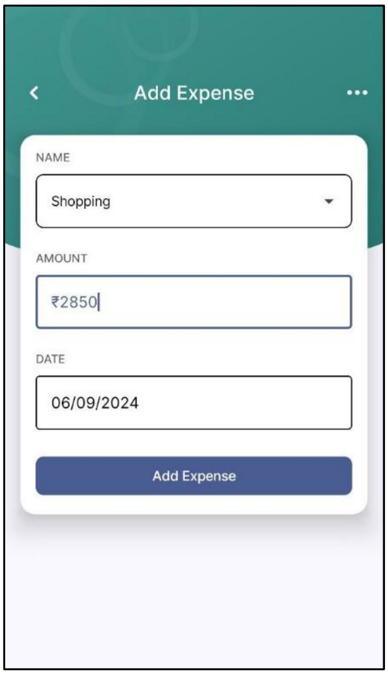
2. Add Transaction:

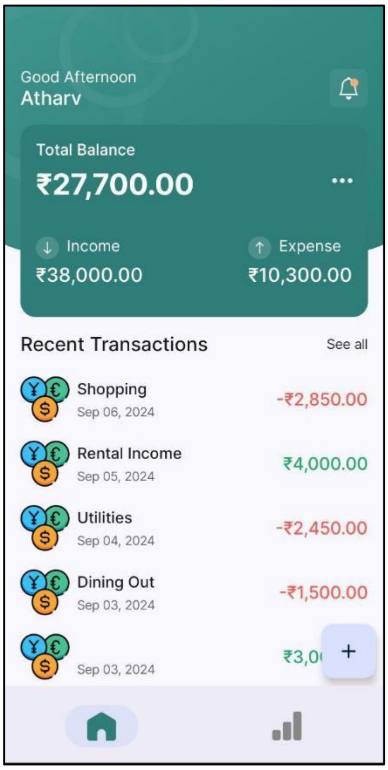
a. Add Income:





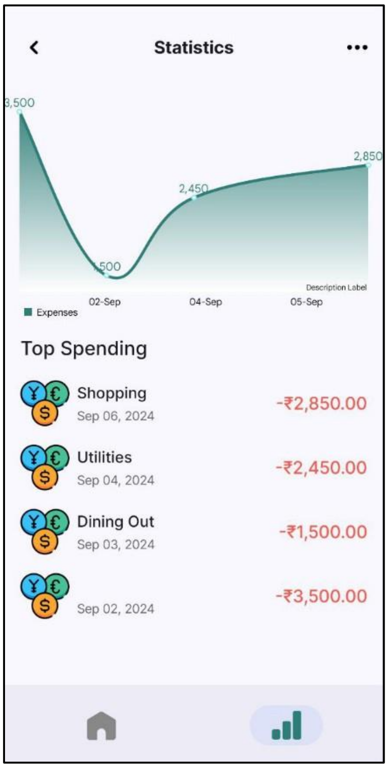
a. **Add expense:**





3. Transaction List:

4. Statistics screen:



Chapter 6: Conclusion

The virtual internship of Eduskills Android Developer was an enriching experience that provided a structured and practical approach to Android development using basins. Through a number of well -defined tasks, I have gained a comprehensive understanding of Android App Architecture, the principles of user interface design, database management and state management techniques. The internship helped to bridge the gap between the theoretical knowledge of my open optional course on the development of Android and the real-world application by allowing me to work on practical projects.

One of the most important aspects of the internship was the Final Assessment project, where I developed Android Tracker Tracker. This project allowed me to integrate all the skills acquired during internship, including the design of the user interface, the ViewModel and Livedata implementation for efficient data management and the use of a database for persistent storage. The key challenge in the project was to ensure user interface updates in real time, which I overcame by efficient use of Android life cycle components. The project also strengthened my understanding of application tuning, performance optimization and the best coding procedures.

In addition to technical skills, this internship instilled basic professional values such as problems, attention to details, adaptability and time management. He also emphasized the importance of continuous learning and cooperation that is critical in the dynamic field, such as software development.

Overall, this internship was a very valuable learning experience that equipped me with the necessary skills and confidence in developing Android applications in the real world. Exposure to industry standard tools and proven practices has contributed significantly to my growth as Android developer and prepared me for future opportunities in the development of mobile applications.

Chapter 7: Bibliography

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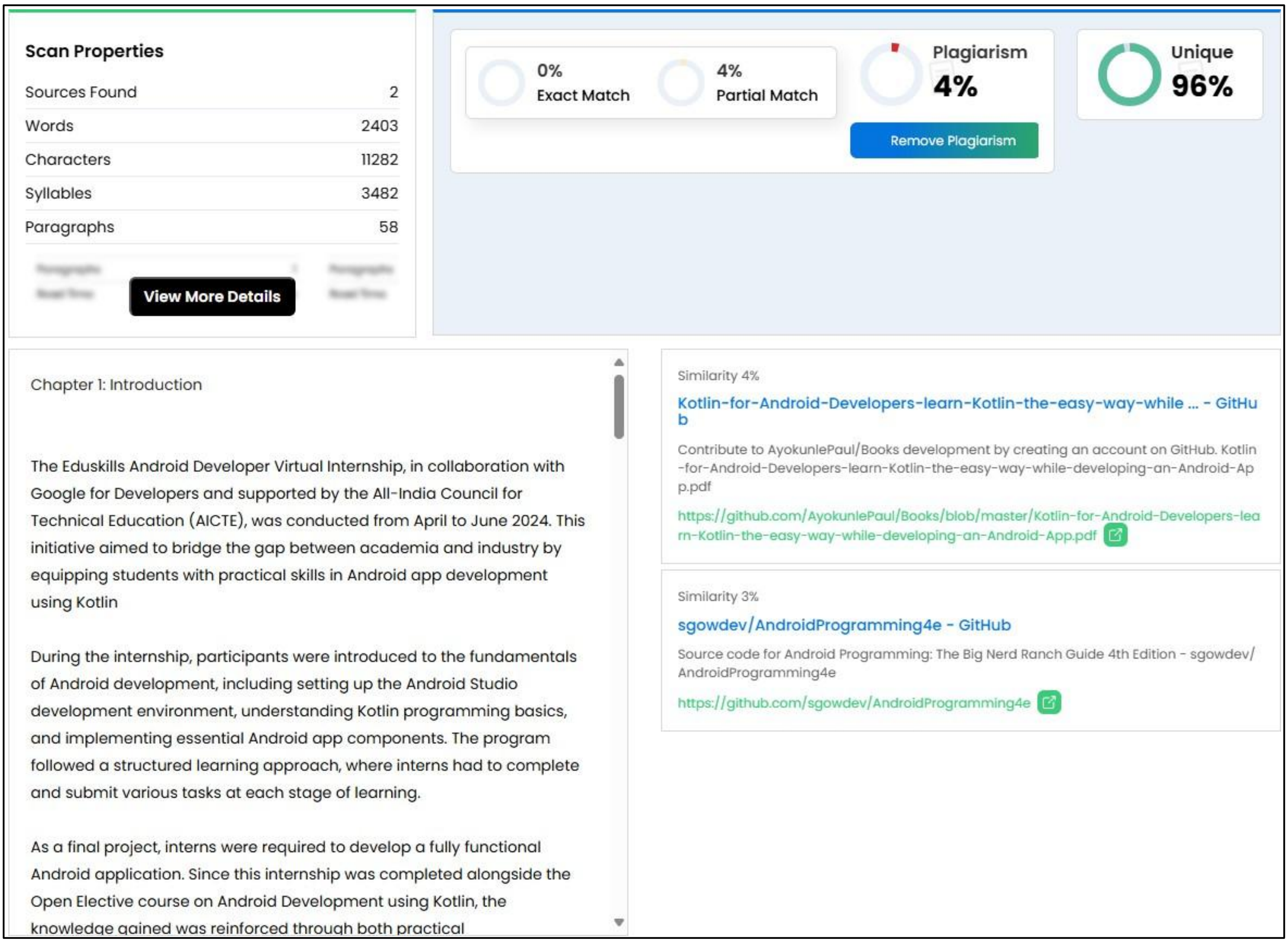
Chapter 8: Plagiarism Report



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Internship Certificate


N·E·A·T
प्रौद्योगिकी के लिए राष्ट्रीय शैक्षणिक सहयोग
National Educational Alliance for Technology

AICTE

EduSkills
Nation Building Through Skills

अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education

AICTE · EduSkills 2024 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Piyush Hemant Hole

Pimpri Chinchwad College of Engineering

has successfully completed 10 weeks

Android Developer Virtual Internship

During April - June 2024

Supported By : India Edu Program

Google for Developers

Karthik Padmanabhan
Developer Ecosystem Lead
MENA & India, Google

Shri Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
NEAT Cell, AICTE

Dr. Satya Ranjan Biswal
Chief Technology Officer (CTO)
EduSkills

Certificate ID :0cc89d1252572b3b745f41c89d0f47f8
Student ID : STU6581a245debc51702994501

INTERNSHIP GRADE B

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30